

Pratyush Padhy

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EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science, Computer Science — GPA: 3.86/4.00

Expected Graduation: 2027

- Dean's Honor List: Fall '25, Winter '26 | **Organizations:** Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI
- **Awards:** California ELC Award (**Top 9%**), AP Scholar with Distinction, **5th Place** @ DECA State (EIP)

PROFESSIONAL EXPERIENCE

Data @ UCI

Irvine, CA

Undergraduate Mentor

Jan. 2026 – Mar. 2026

- Mentored **50+** students on data-focused projects and dataset extraction using **Python**, **HTML**, and **JavaScript**.
- Guided mentees in data collection, variable selection, and exploratory analysis using **Kaggle** datasets.
- Supported predictive model development, training, and evaluation using **Google Colab** and GitHub Repositories.
- Provided feedback and evaluated final project presentations conducted at the Data @ UCI mentorship meeting.

Computers 4 Kids

Sacramento, CA

Software Intern

Aug. 2023 – May 2025

- Executed collection and refurbishment of computers for underrepresented students, addressing inequitable tech access.
- Collaborated with leadership to develop scalable strategies tackling the digital divide in underserved communities.
- Established a school-based chapter with **200+** members, expanding the organization's digital access programs.
- Coordinated computer distribution events, ensuring refurbished devices met quality standards for students.

Robotics for All

Palo Alto, CA

Software Intern

Jun. 2023 – Aug. 2024

- Deployed an asynchronous training system with readiness assessments for **24+** instructors, improving preparedness.
- Conducted interviews with incoming volunteers to evaluate teaching readiness and provide targeted feedback.
- Coached a coding class of **20** students; built a parent feedback questionnaire to improve curriculum.
- Built an AI-powered chatbot for organization website; awarded the Presidential Service Award (Gold) for **250+** hours.

PROJECTS

FacePulse | Python, TensorFlow, FastAPI, React, OpenCV, Dlib | face-pulse.netlify.app

Jan. 2026 – Present

- **Featured on Handshake AI Showcase**, selected among thousands of student projects for national recognition.
- Evolved from a facial landmark emotion detector built at Inspirit AI into a full comparative study of **MLP**, **CNN**, and **VGG16** architectures on **FER2013** (**35,000+** images), achieving **93%** accuracy with **2-phase** VGG16 fine-tuning.
- Built a full-stack web app with a **FastAPI** + **TensorFlow** backend deployed via **Docker** on **Hugging Face Spaces** and a **React/TypeScript** frontend on Netlify with real-time webcam emotion detection.
- Implemented a session analytics dashboard with emotion probability charts, session timeline, and animated transitions using Chart.js and **Framer Motion**.

Berkeley ROAR | Python, NumPy, Pandas, Matplotlib, Machine Learning

Jun. 2023 – Aug. 2023

- Trained under UC Berkeley faculty in **Python**, artificial intelligence, autonomous systems, and optimization.
- Optimized a Tesla Model 3 simulation using machine learning, neural networks, and decision trees in **Python**.
- Designed and tuned waypoint-based speed control strategies to improve simulation lap times, leading the team to place in the **top 10**.

iLab Stanford — RNA-seq Research | R, IGV, Bioinformatics, Statistical Analysis

Dec. 2023 – Jul. 2024

- Analyzed human bulk RNA-seq data for non-small cell lung cancer (**GSE268175**), mapping **UQCR11** and **OAZ1** genome expression across chromosomes using **IGV** for visualization.
- Generated dimensionality heatmaps and statistical plots in **R** for single-cell RNA sequencing analysis; presented research findings to Dr. Qian Wang of Stanford's iLab.

TECHNICAL SKILLS

Languages: Python, Java, R, C++, HTML, CSS, JavaScript, TypeScript

Libraries & Frameworks: TensorFlow, Keras, OpenCV, Dlib, NumPy, Pandas, Matplotlib, scikit-learn, Seaborn, Chart.js, React, Vite, FastAPI, Express.js, Tailwind CSS

Tools & Platforms: Git, VS Code, PyCharm, Google Colab, Docker, Hugging Face Spaces, Data Analysis in R

Certifications: PCEP – Python, Coursera UCSC – C++, WorldTutor – Java